

# **DAT655 Blockchain technology and Applications**

**Intro**

**Leander Jehl**

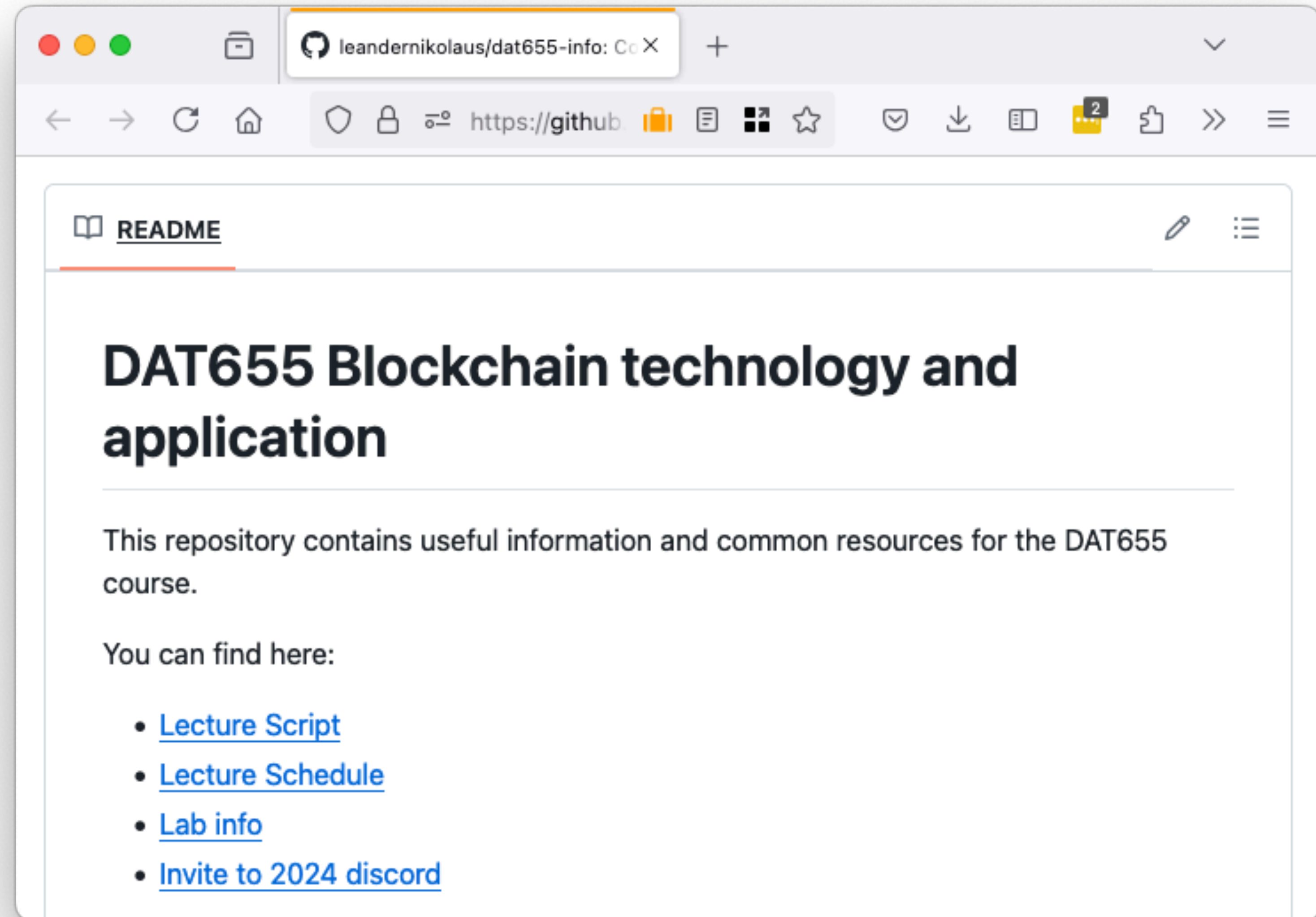
# About me

Leander



# Course info

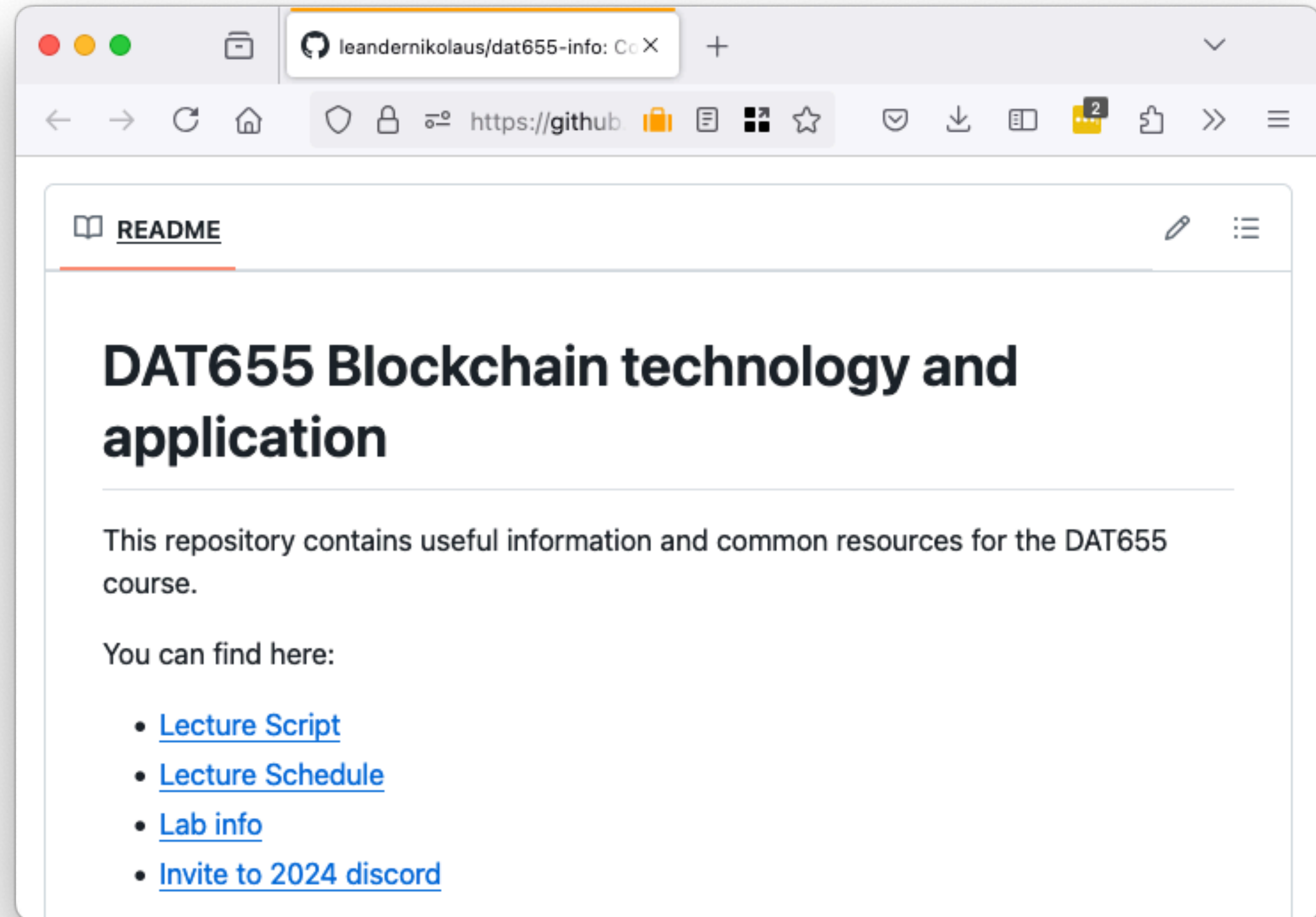
<https://github.com/leandernikolaus/dat655-info/>



# Course info

<https://github.com/leandernikolaus/dat655-info/>

- Lectures
- Reading material
- Lecture script  
(will be updated)
- Slides  
(will be updated)



# Course info

<https://github.com/leandernikolaus/dat655-info/>

- Lectures
  - Mainly theory
- Labs
  - Coding and discussion

## Weekly timetable

| Day     | Time          |         | Room      |
|---------|---------------|---------|-----------|
| Monday  | 10:15 - 12:00 | Lab     | EAL H-211 |
| Tuesday | 12:15 - 14:00 | Lecture | KE A-203  |

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- Timetable
  - Topic for Tuesday lecture
  - Monday lab activity:
    - Coding  
Live and exercise
    - Discussion

## Week 35 Lecture 1

*Intro and Hash based data structure*

- [Info slides](#)
- [Slides](#)
- [Script](#) Chapter 1

### Lab

- [Notebook](#)

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- Lab
  - Together or alone
    - Attend lab or deliver your own solution
  - Attending counts as passing the assignment

| Week | Lecture | Topic                     | Lab/Discussion                    |
|------|---------|---------------------------|-----------------------------------|
| 34   | Leander | Intro and basic           | Lab (Notebook)                    |
| 35   | Leander | Proof of Work             | Lab (Notebook)                    |
| 36   | Leander | Accounts and UTXO         | Lab (Notebook)                    |
| 37   | Leander | TBA                       | Lab (Notebook)                    |
| 38   | Leander | Applications              | Discussion (Applications)         |
| 39   | Leander | To PoS and Committees     | Lab (Notebook)                    |
| 40   | Leander | Ethereum                  | Discussion (Decentralization)     |
| 41   | Leander | SmartContracts (security) | Solidity                          |
| 42   | Leander | TBA                       | Solidity and tokens               |
| 43   | Leander | Layer 2                   | Discussion (Layer 2 applications) |
| 44   | Leander | Privacy in Blockchain     | Discussion (Privacy pro and con)  |
| 45   | Leander | DePIN & DID               | Discussion (Token applications)   |
| 46   | Leander | Presentations             | Presentations                     |

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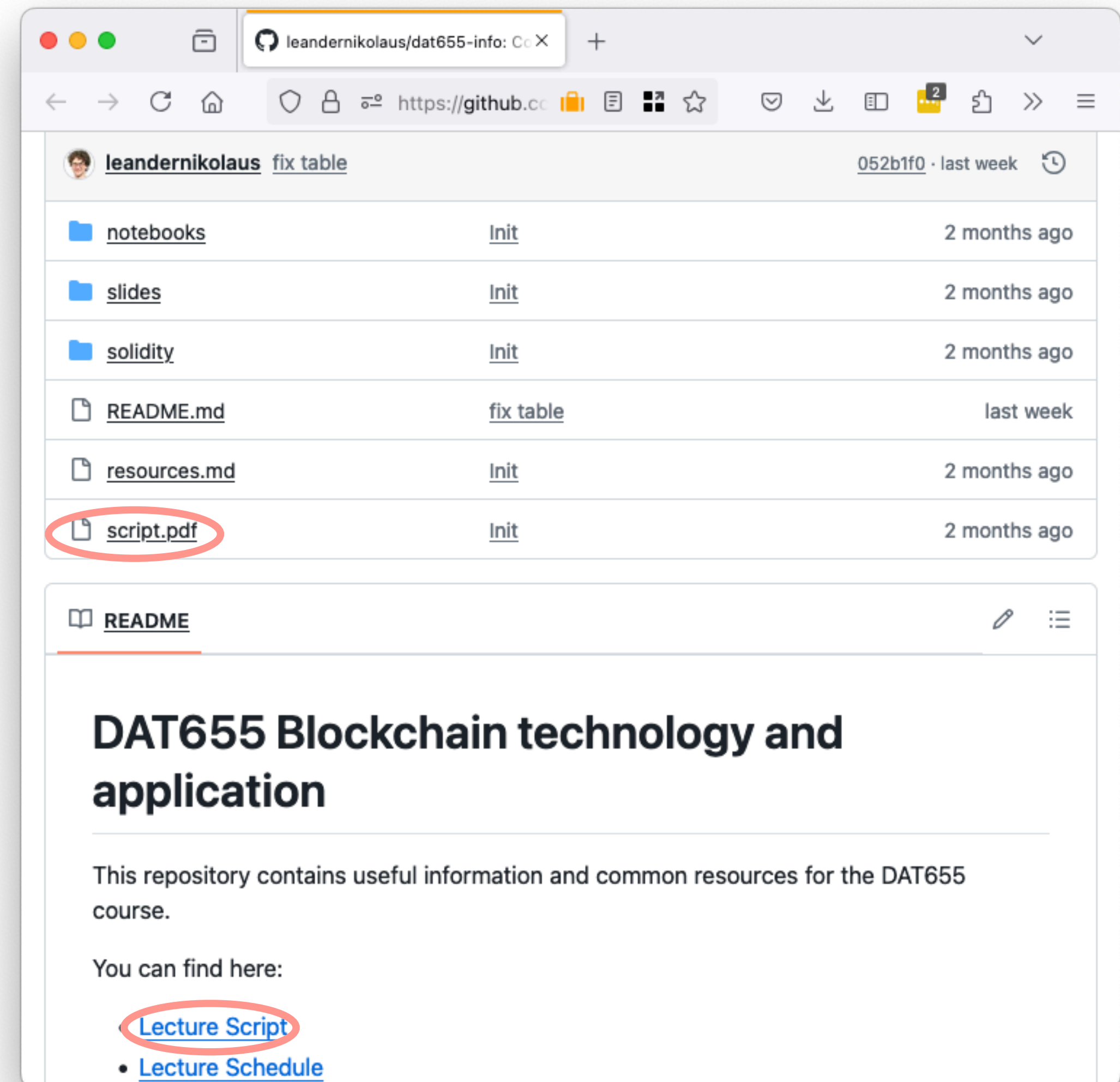
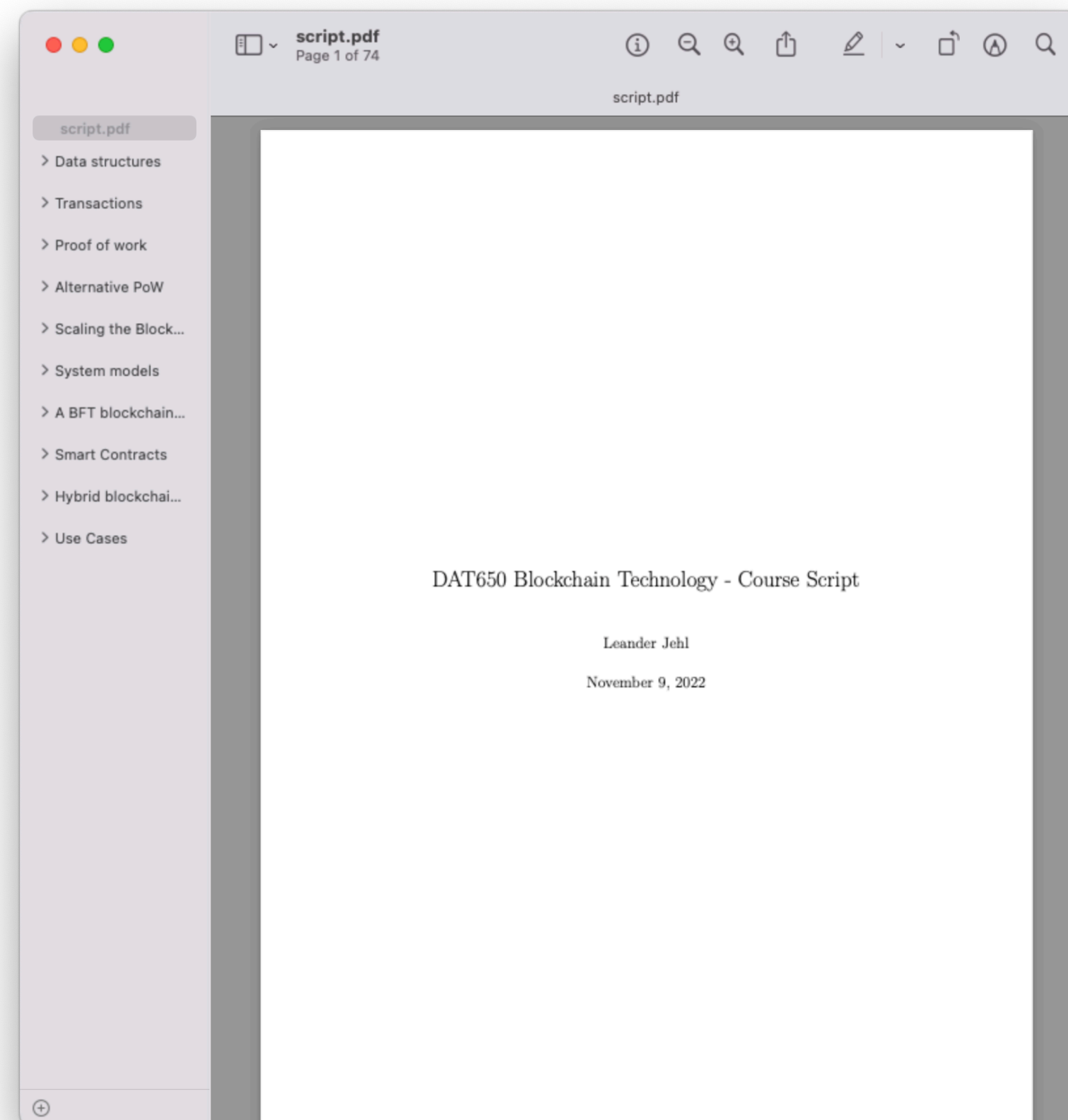
- Discussions
  - Attend or presend
    - Participate in discussions
- or
- Present a use case based on a review paper

| Week | Lecture | Topic                     | Lab/Discussion                    |
|------|---------|---------------------------|-----------------------------------|
| 34   | Leander | Intro and basic           | Lab (Notebook)                    |
| 35   | Leander | Proof of Work             | Lab (Notebook)                    |
| 36   | Leander | Accounts and UTXO         | Lab (Notebook)                    |
| 37   | Leander | TBA                       | Lab (Notebook)                    |
| 38   | Leander | Applications              | Discussion (Applications)         |
| 39   | Leander | To PoS and Committees     | Lab (Notebook)                    |
| 40   | Leander | Ethereum                  | Discussion (Decentralization)     |
| 41   | Leander | SmartContracts (security) | Solidity                          |
| 42   | Leander | TBA                       | Solidity and tokens               |
| 43   | Leander | Layer 2                   | Discussion (Layer 2 applications) |
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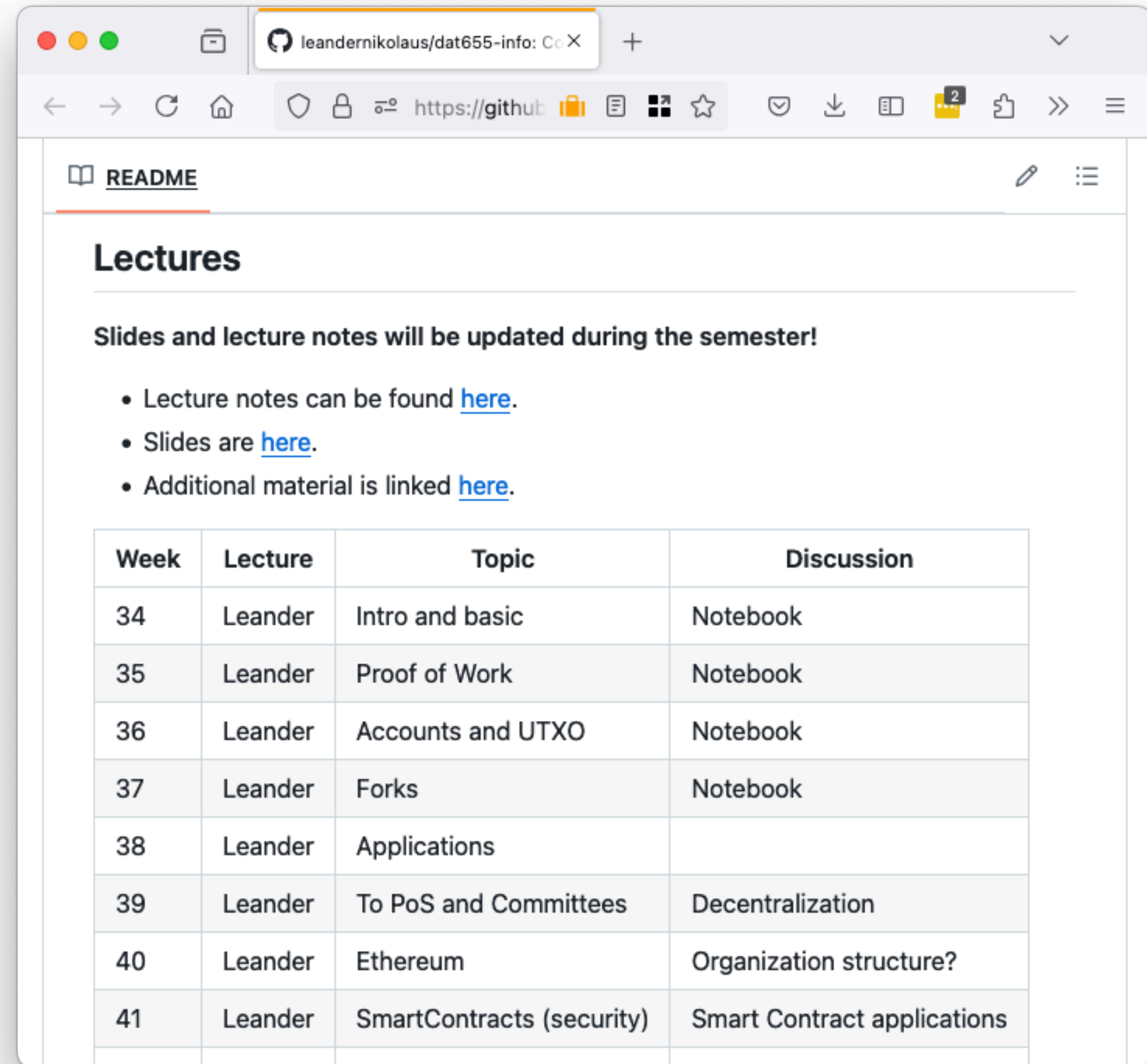
<https://github.com/leandernikolaus/dat655-info/>

- Lecture script



# Grading

- Labs approved
- Participation or Presentation
- Oral exam covering
  - Lectures
  - Labs
  - Discussions



The screenshot shows a web browser window displaying a GitHub README page for the repository 'leandernikolaus/dat655-info'. The page title is 'README'. The main heading is 'Lectures'. Below the heading, a message states: 'Slides and lecture notes will be updated during the semester!'. This is followed by a bulleted list: 'Lecture notes can be found [here](#).', 'Slides are [here](#).', and 'Additional material is linked [here](#)'. Below the list is a table with four columns: 'Week', 'Lecture', 'Topic', and 'Discussion'. The table contains data for weeks 34 through 41.

| Week | Lecture | Topic                     | Discussion                  |
|------|---------|---------------------------|-----------------------------|
| 34   | Leander | Intro and basic           | Notebook                    |
| 35   | Leander | Proof of Work             | Notebook                    |
| 36   | Leander | Accounts and UTXO         | Notebook                    |
| 37   | Leander | Forks                     | Notebook                    |
| 38   | Leander | Applications              |                             |
| 39   | Leander | To PoS and Committees     | Decentralization            |
| 40   | Leander | Ethereum                  | Organization structure?     |
| 41   | Leander | SmartContracts (security) | Smart Contract applications |